Confirming Memory Leaks on a Live Website

Shivam is sitting in an interview and gives LinkedIn as an example.

He checked for memory leaks on the LinkedIn website using Chrome DevTools Heap Snapshots.

He tested this by performing normal LinkedIn actions such as:

• Scrolling the feed
• Opening profiles
• Navigating between pages
• Opening modals
• Using the search feature

1) Check Object Types

First, I checked if the same object names appear in both snapshots.

Examples:

- Detached <input>
- PerformanceEntry
- MutationObserver

Why this is important:

- These objects are created by the browser and frameworks
- Seeing them again is normal

Same objects in both snapshots → Normal behavior

2) Compare Object Count (Most Important)

Then I checked how many times each object appears.

Example:

- Detached <input> → 10 → 10
- PerformanceEntry → 19 → 19

Why this is important:

- Memory leaks show numbers that keep growing
- Stable numbers mean memory is released

Numbers did not increase. Memory leaks happen only when these numbers keep increasing.

3) Compare Memory Size

Checked if memory size changed.

Example:

- 0.3 kB → 0.3 kB

Why this is important:

- If memory is released, size stays the same
- Leaks keep memory in use

Memory is stable
A leak looks like: 0.3 kB → 2.5 MB

4) Check Percentage

All objects showed:
0% growth

Why this is important:

- Growth % shows if memory is increasing
- 0% means no extra memory is used

Memory is not growing

5) Check Detached Elements

Important rule:

1. Detached elements are normal
2. Detached elements that increase = memory leak

Why detached elements happen:

- Page re-renders
- UI updates
- Old DOM removed correctly

Result:

- Detached <input> → 10 → 10
- Detached <body> → 17 → 17

Stable → No memory leak

How to Decide a Memory Leak

No memory leak when:

- Object count stays the same

- Memory size stays the same
- Detached elements don’t grow

Memory leak when:

- Object count increases after actions
- Detached elements keep adding
- Memory does not go down after GC

Correct Way to Test

1. Take Snapshot A
2. Do some action (scroll, navigate, open modal)
3. Force Garbage Collection
4. Take Snapshot B
5. Repeat

Numbers increase → Memory leak
Numbers stay same → Safe

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